

STYLE-CONDITIONED VIDEO CAPTIONING

TextSink

Four voices. One truth.

FORMAL

SARCASTIC

TECH-HUMOR

EVERYDAY-HUMOR

Grounded on frames + audio, styled by Gemma 4 — faithful and on-tone.

AMD Developer Hackathon: ACT II · Track 2 | Gemma 4 · Fireworks AI · AMD

Two jobs. Most entries do one.

Track 2 scores every caption on accuracy AND tone — across four different styles.

Accurate but flat

Frame-only bots describe the scene correctly, then miss the joke — and miss everything that was spoken.

Funny but wrong

Prompt-only bots write a punchy line — that invents objects, people, or events never shown on screen.

TextSink is built to win both axes at once.

One clip in. Four voices out.

A single 30s–2min clip becomes four captions — each faithful to the same facts.

● FORMAL

Neutral, precise, broadcast register.

● SARCASTIC

Dry and deadpan — never mean.

● TECH-HUMOR

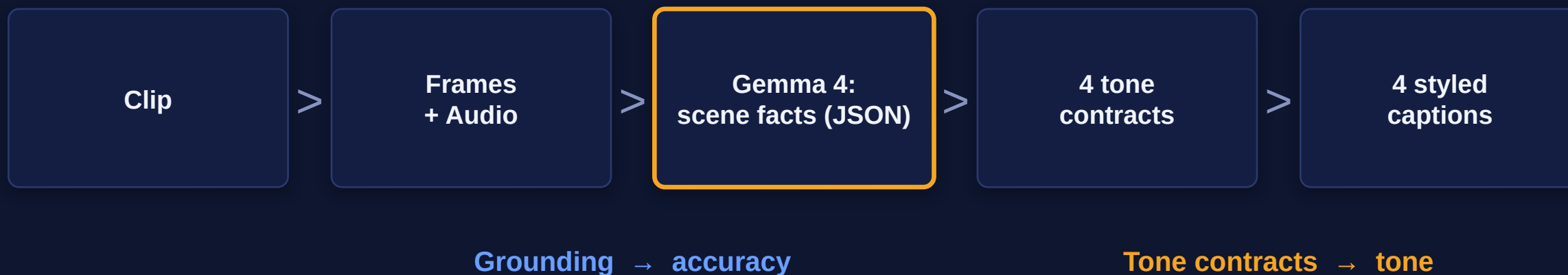
A joke a developer actually gets.

● EVERYDAY-HUMOR

Relatable, warm, zero jargon.

How it works

Separate the two things the judge scores — then optimize each.



One model — Gemma 4 — does the grounding AND the four captions. ffmpeg + Whisper feed it; per-style contracts shape the voice.

Accuracy: ground before you write.

- Sample frames every ~5s with ffmpeg (8–16 per clip).
- Pull audio into a transcript with Whisper — catches spoken content frame-only bots miss.
- Gemma extracts only what is visible or audible, as structured facts.
- Every caption is locked to those facts. No invented objects, people, or brands.

scene facts (grounded)

```
{  
  "setting": "indoor room, daytime",  
  "subjects": ["a person", "a laptop"],  
  "actions": ["types", "smiles at  
              screen"],  
  "notable_audio": ["upbeat music"],  
  "mood": "focused"  
}
```

Tone: four contracts, four voices.

Same grounded scene. Four Gemma 4 captions — real, verified output.

SCENE official judge clip: an orange tabby kitten walks toward the camera through sunlit foliage

FORMAL

"An orange tabby kitten walks forward through the foliage toward the camera in a wooded area."

SARCASTIC

"A fierce predator emerges from the brush, clearly ready to conquer the entire forest one tiny, uncoordinated step at a time."

TECH-HUMOR

"The latest AI agent successfully navigating its first training environment. It's small, but the feature set is looking promising."

EVERYDAY-HUMOR

"A tiny orange explorer makes a grand entrance, bravely navigating the treacherous jungle of backyard leaves."

Gemma 4 — end to end.

\$5,500

addressable with one project:

\$2,500 Track 2 + \$3,000 Best Use of Gemma
4

- One model, Gemma 4 (26B MoE), does BOTH the visual grounding and the four styled captions.
- Served on-demand via Fireworks (H200), scale-to-zero — only bills while captioning.
- MIT-licensed, single Docker container — clone, build, run.
- Qwen 3.7 Plus + gpt-oss serverless fallback for zero-setup dev.

Runs today — not a mockup.

- Gemma 4 deployment verified: accurate grounding + 4 clean, on-tone captions.
- Real ffmpeg frame sampling and audio extraction verified.
- Full pipeline runs end-to-end and emits valid 4-style JSON; schema checks pass.
- Ships a self-judge that scores accuracy + tone to tune prompts with data.

run it

```
docker run --rm \
  -e FIREWORKS_API_KEY=$KEY \
  -v $PWD/input:/input:ro \
  -v $PWD/out:/output \
  textsink
```

Four voices. One truth.

github.com/banksythequantLab/textsink